Module 4 Assignment: Risk Mitigation

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ALY6130: Risk Management for Analytics

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Introduction and Rationale for Quantitative Analysis

Risk #	The Risk of/That	Likelihood	Impact Sco
1	Infrastructure upgrade costs for logistics technology	3	2
2	Vendor switching costs for logistics service providers	3	4
3	Logistics operating costs in human capital and warehouse infrastructure	5	6
4	Margin reduction due to price competition	9	8
5	Capital investment in logistics infrastructure	5	9
6	Process adjustments in fulfillment workflow	3	1
7	Limited human capital to support logistics operations	5	2
8	Third-party warehouse and delivery partner dependency	5	4
9	Integration between logistics systems and e-commerce platform	7	6
10	Fulfillment network disruption during peak demand periods	7	8
11	System failure or fulfillment network downtime	3	9
12	Limited competitive pressure from small platforms	3	1
13	Competitor promotional campaigns	5	2
14	Competitors introducing improved fulfillment capabilities	5	4
15	Competitors offering faster or lower-cost fulfillment services	7	6
16	Cost of lagging behind competitors with advanced logistics networks	7	8
17	Strong market competition from platforms with mature logistics networks	9	9
18	End-to-end commerce solution improving user experience	7	1
19	Dissatisfaction due to delivery delays	5	4
20	Logistic system and service failure affecting user experience	5	8
21	Data privacy breaches affecting user trust	5	9
22	Platform maintenance and system updates	3	2
23	System scalability limitations during high order volumes	7	4
24	Data integration across fulfillment platforms	5	6
25	Platform performance failure affecting order processing	5	8
26	Large-scale logistics platform outage	1	9
27	Regulatory reporting requirements for logistics operations	3	1
28	Contract compliance review with logistics partners	3	2
29	Transparency requirements for logistics data reporting	3	4
30	Contract disputes with logistics providers	3	6
31	Regulatory changes affecting cross-border shipping	5	8
32	Compliance risk related to customer data and shipping regulations	3	9

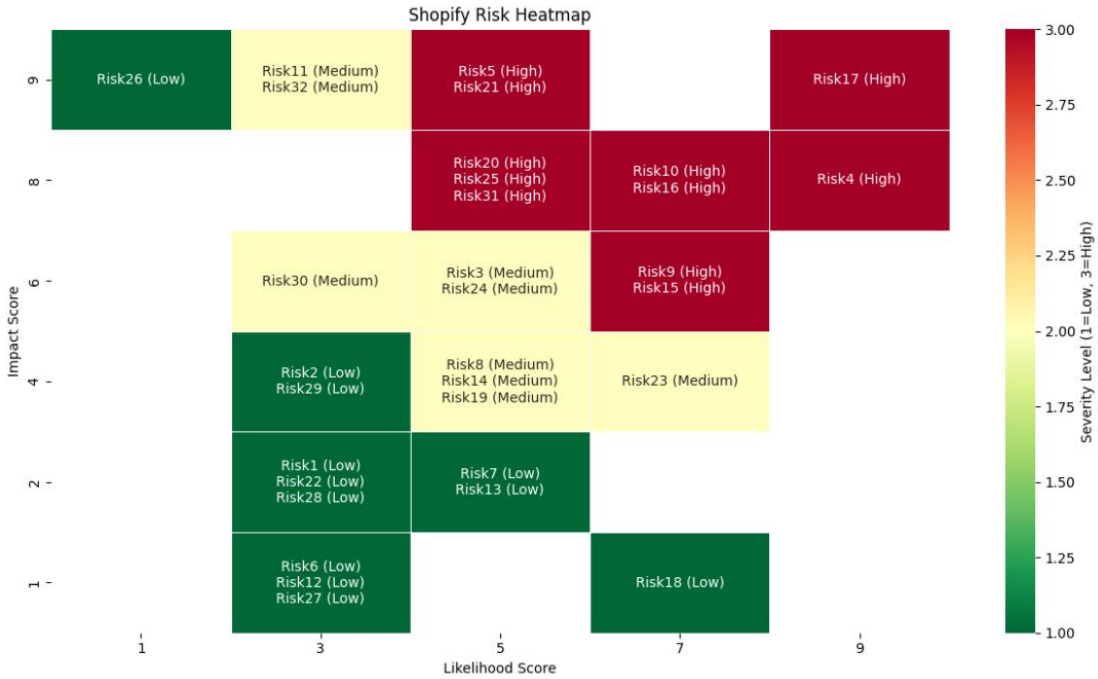
As we learned in Module 3, the first step in understanding risk is to use qualitative methods. This means we identify possible risks and then judge how likely they are and how much damage they could cause based on our knowledge and experience. For Shopify's acquisition of the logistics company Deliverr, we followed this process carefully. We reviewed Shopify's business model, its plans for logistics, and the competitive environment where it operates. We also looked at outside sources such as news articles and industry reports to understand the challenges other companies have faced when building delivery networks. Through this process, we identified 32 separate risks across six different categories, including financial, operational, competitive, reputational, technical, and legal areas. Each risk received a likelihood score from 1 to 9 and an impact score from 1 to 9. We multiplied these scores to get a total risk score and used these numbers to create heatmaps that show which risks are most dangerous.

The heatmaps clearly show that three risks sit in the top right corner where likelihood and impact are both high. These are Risk 17, strong market competition from platforms with mature logistics networks, with a score of 81. Risk 4, margin reduction due to price competition, with a score of 72. And Risk 10, fulfillment network disruption during peak demand periods, with a score of 56. While the qualitative method told us these are our biggest concerns, it did not tell us exactly how much money might be at risk or the precise chance of these events happening. To get this deeper understanding, we must now apply quantitative risk analytics using the **Indicators and Warnings** technique to these three priority risks.

Understanding the Indicators and Warnings Method

The Indicators and Warnings method is a way to turn qualitative concerns into measurable numbers that we can track over time. The basic idea is simple but powerful. For each risk, we identify specific things we can count or measure that would tell us the risk is getting worse. These are our indicators. For each indicator, we set two threshold levels. The first threshold is our warning level. When an indicator crosses this line, it means we should pay closer attention and begin preparing for possible action. The second threshold is our action level. When an indicator crosses this line, it means the risk is happening and we must respond immediately. After we set up these indicators and thresholds, we then use quantitative methods to determine two important things. The first is the probability that our indicators will cross the action threshold within a certain time period, such as the next 12 months. The second is the financial impact on the company if that action level is reached. By multiplying these two numbers together, we get a quantified risk exposure. This number represents the probable cost of the risk and helps us decide how much money and effort we should spend to prevent it. For this assessment, we would use two main quantitative techniques. Monte Carlo simulation would help us model the probability of different events by running thousands of possible future scenarios based on past data and reasonable assumptions. Machine learning methods would help us find patterns in large amounts of data and make more accurate predictions about future performance, especially for complex operational risks like network failures during busy periods.

Quantitative Assessment of Strong Market Competition



Risk number 17, strong market competition from platforms with mature logistics networks, received the highest possible qualitative score of 81 with likelihood 9 and impact 9. This tells us the team believes it is very likely to happen and would cause

extremely high damage. To assess this risk quantitatively using the I&W method, we would first identify three measurable indicators. The first indicator is the rate at which larger merchants begin moving more of their sales to competing fulfillment networks like Amazon FBA. We would measure this monthly by analyzing sales data from our top merchants. The second indicator compares shipping speed between Shopify's network and Amazon's network in major city areas. We would measure the average delivery time for two-day shipping promises in cities like New York, Los Angeles, and Chicago. The third indicator measures whether Shopify keeps the same share of its merchants' total sales over time. We would calculate this by looking at the gross merchandise value processed through Shopify compared to the total sales of our top 1000 merchants across all platforms. For each indicator, we would set warning and action thresholds based on historical data and industry benchmarks. For the merchant attrition indicator, a monthly rate above 2 percent would trigger a warning, and a rate above 5 percent would trigger action. For the shipping speed indicator, if Shopify falls one day behind on average, we would issue a warning, and if it falls two or more days behind, we would take action. For the share of wallet indicator, a yearly drop of 3 percent would trigger a warning, and a drop of 7 percent would trigger action. To determine the probability of hitting these action thresholds, we would gather past data on merchant behavior, competitor shipping performance, and market trends over the last three to five years. We would then build a Monte Carlo simulation that runs thousands of possible future scenarios. This simulation would consider factors like the historical volatility of shipping performance, the rate at which competitors have improved their services in the past, and how merchants have responded to these changes. The simulation would output the probability of hitting each action threshold within the next 12 months. To calculate the financial impact, we would analyze what happens when merchants leave the platform. If 5 percent of our top merchants move significant volume to competitors, we would lose their subscription fees and the transaction processing revenue from their sales. Based on current average revenue per merchant, this could amount to approximately 100 million dollars annually. If our Monte Carlo simulation shows a 20 percent probability of hitting this action threshold within a year, then our quantified risk exposure for Risk 17 would be 20 million dollars. This number comes from multiplying the 20 percent probability by the 100 million dollar impact.

Quantitative Assessment of Margin Reduction

Risk number 4, margin reduction due to price competition, received a qualitative score of 72 with likelihood 9 and impact 8. This means the team believes it is very likely to happen and would cause serious damage to company profits. To assess this risk quantitatively, we would identify two main indicators. The first indicator is the gap between what Shopify charges merchants for delivery and what Amazon charges for similar services. We would collect pricing data from both companies on a regular basis and calculate the average percentage difference. The second indicator tracks how much money the company spends on marketing campaigns that promote shipping deals, measured quarterly against the budget. For the pricing gap indicator, if

Shopify's price is less than 5 percent higher than Amazon's, we would issue a warning because the gap is getting too small. If Shopify must match Amazon's price exactly or even charge less, this would trigger action because it would require the company to lose money on each delivery. For the marketing spend indicator, going over budget by 5 million dollars in a quarter would trigger a warning, and going over by 15 million dollars would trigger action. To determine the probability of hitting the action threshold for pricing, we would analyze the competitive landscape. We might ask, what is the chance that Amazon reduces its fulfillment fees by 10 percent in the next six months? This probability could come from market analysis, expert opinion, or historical patterns of competitor behavior. If we determine there is a 40 percent chance that such a price change will force Shopify to match and subsidize deliveries, we would then calculate the financial impact. If Shopify must subsidize 10 million deliveries by 2 dollars each to remain competitive, the quarterly impact would be 20 million dollars. Multiplying the 40 percent probability by the 20 million dollar cost gives us a quantified risk exposure of 8 million dollars for the quarter.

Quantitative Assessment of Fulfillment Network Disruption

Risk number 10, fulfillment network disruption during peak demand periods, received a qualitative score of 56 with likelihood 7 and impact 8. This means the team believes it is somewhat likely to happen and would cause high damage. To assess this risk quantitatively, we would identify three indicators. The first indicator is order processing latency, measured as the time from order placement to hand-off to the carrier. The second indicator is the on-time delivery rate for Shop Promise orders. The third indicator is the system error rate, measured as the number of 5xx errors or failed API calls between the Shopify platform and fulfillment centers. For the latency indicator, if processing takes longer than four hours for more than 1 percent of orders, we would issue a warning. If it takes longer than twelve hours for more than 5 percent of orders, we would take action. For the on-time delivery indicator, a rate below 95 percent would trigger a warning, and a rate below 85 percent would trigger action. For the error rate indicator, a rate above 0.5 percent would trigger a warning, and a rate above 2 percent would trigger action. To determine the probability of hitting these action thresholds during peak periods like Black Friday and Cyber Monday, we would use machine learning techniques. We would train a time-series forecasting model on historical data showing order volume, system latency, error rates, and on-time performance from previous peak seasons. This model would learn patterns and relationships and could predict the likelihood of hitting each threshold on any given day during the upcoming busy period. If this model showed a 5 percent chance of a major failure on the busiest shopping day, we would then calculate the financial impact. Such a failure would include direct costs like refunds to merchants for late deliveries and credits for missed Shop Promise guarantees. It would also include indirect costs like lost future sales because merchants lose trust in the platform. Based on past service disruptions and their effect on gross merchandise value, we might estimate the total impact at 200 million dollars. Multiplying the 5 percent probability by the 200 million dollar cost gives us a quantified risk exposure of 10 million dollars

for that single day.

Updating the Risk Treatment Plan and Next Steps

After completing this quantitative assessment, we would update the Risk Treatment Plan and Risk Register with the summarized high-level results. For Risk 17, we would note a quantified exposure of 20 million dollars annually based on a 20 percent probability of significant merchant attrition. For Risk 4, we would note a quarterly exposure of 8 million dollars based on a 40 percent probability of a price-matching scenario. For Risk 10, we would note a single-day exposure of 10 million dollars for the peak shopping period based on a 5 percent probability of major disruption. These numbers give us clear targets for mitigation investments. We would recommend specific actions for each risk, such as building systems to track our indicators in real time, developing response plans that can be activated quickly when warning thresholds are crossed, and adding extra computer capacity before the holiday season. The Python notebook appended to this report contains the foundational code for the Monte Carlo simulations and machine learning models described in this analysis, along with the data sources used. By following this approach, we turn risk management from a simple paper exercise into a practical tool for protecting the company's future based on real data rather than just feelings or guesses.

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